

SIMATS ENGINEERING

Department of Computer Science and Engineering



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CSA15 Cloud Computing and Big Data Analytics

CO2 AT1 - Capstone Infrastructure Sizing Workbook

Guided calibration workbook for VM sizing, virtualization decisions, snapshot planning and VM-container comparison

Assessment Metadata

Course Code	CSA1517
Course Title	Cloud Computing and Big Data Analytics
Assessment	CO2 AT1 - Virtualization Scenario Problem-Solving
Submission Type	Individual digital workbook based on the same capstone title used in CO1
Faculty	Dr. C. Rajesh Babu
Employee ID	SSETSCS-415
Submission Mode	Completed workbook and evidence repository link through Google Classroom

How This Workbook Works

- Read each guidance block and immediately fill the related response table.
- Use the same capstone title selected for CO1.
- Make reasonable assumptions when exact values are unknown. Good assumptions must be written clearly.
- The goal is not to guess a perfect industry configuration; the goal is to reason technically and justify the first infrastructure plan.
- The final decisions from this workbook will be used again in CO2 AT2 and CO2 AT3.

Technical Tip

A good infrastructure estimate is transparent. If a number is assumed, write why it is assumed. If a number is calculated, show the formula. If a service is selected, justify why it fits the workload.

Student and Capstone Details

Student Name	VALLAPU MOHAN KUMAR
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Class / Slot	Slot C

Capstone Title	Food Trace Cloud Supply Chain Tracker with QR Provenance and Spoilage Prediction
Capstone Product Type	Web and Mobile Application (Hybrid App)
CO1 Architecture Reference	Mention concept map / presentation title / GitHub link if available

Activity 1: Bring Forward CO1 Capstone Context

Start with the cloud product idea already used in CO1. The infrastructure plan must support that same application, not a new unrelated example.

Capstone Element	Student Entry
Primary users	Farmers, distributors, retailers, customers, administrators
User roles	Admin manages system, suppliers upload products, customers scan QR codes
Main modules	Login, QR Code, Product Tracking, Spoilage Prediction, Dashboard
Cloud services identified in CO1	SaaS, PaaS, Firebase, Cloud Storage, REST API
API/backend need	REST API for communication between application and cloud
Database need	Firebase Firestore Database
File upload/storage need	Cloud Storage for product images and QR codes
Analytics/AI/Python module	Python-based spoilage prediction and analytics

Activity 2: Workload Classification

Classify the capstone workload before selecting a VM. A form-based SaaS app and an AI analytics app should not receive the same infrastructure plan.

Workload Type	Typical Signals	Starting VM Guidance
Light web / form app	Login, CRUD, simple reports, low file upload	1-2 vCPU, 2-4 GB RAM
API backend	Authentication, REST APIs, moderate concurrent requests	2 vCPU, 4-8 GB RAM
Database-heavy app	Search, many records, frequent writes	2-4 vCPU, 8-16 GB RAM
File-heavy app	Images, PDFs, uploads, downloads	2 vCPU, 4-8 GB RAM + separate object storage
Analytics app	Dashboards, aggregation, batch reports	4 vCPU, 16 GB RAM
AI/Python module	Prediction, NLP, image/text processing	4-8 vCPU, 16-32 GB RAM; GPU optional
High traffic app	Large user base and peak demand	Multiple VMs or autoscaling group

Technical Tip

Start small for prototype, but do not hide growth. A valid answer can say: prototype uses one VM, pilot separates database, production uses load balancing and managed services.

Student Decision: My capstone workload type and reason

Workload Type:

API Backend + Database Heavy + Analytics Application

Reason:

The application processes QR scans, stores product information, predicts spoilage, and generates reports using cloud analytics.

Activity 3: User and Traffic Calibration

Use the following formulas to convert user assumptions into approximate traffic.

Metric	Formula	Example
Concurrent Active Users	Registered Users x Active User % x Peak Factor	500 x 10% x 2 = 100
Requests per Minute	Concurrent Active Users x Actions per User per Minute	100 x 2 = 200
Requests per Second	Requests per Minute / 60	200 / 60 = 3.33 RPS
Daily Requests	Daily Active Users x Avg Actions per Day	200 x 30 = 6000

Input	Value	Reason / Assumption
Registered users expected	500	Prototype application with limited users
Active user percentage	10%	Around 10% users active daily
Peak factor	2	Peak-time usage doubles active users
Actions per user per minute	2	QR scan and product tracking
Calculated RPS	3.33	Calculated as 200 ÷ 60 requests per second.
Daily active users	200 Users	Estimated active users accessing the application each day.
Daily requests	6000 Requests (200 × 30)	Calculated as 200 × 30 average actions per day.

Activity 4: CPU Calibration

CPU sizing depends on request rate and processing complexity. Use the score below as a guided estimate.

CPU Driver	Score
Base web/API application	1
Authentication and role management	+1
Moderate API traffic above 5 RPS	+1
Database-heavy search or reporting	+1 to +2
Background jobs or notifications	+1
Analytics or dashboards	+2
AI/Python inference inside server	+2 to +4
Total CPU Score	Recommended vCPU
1-2	1-2 vCPU
3-4	2 vCPU
5-6	4 vCPU
7-9	4-8 vCPU or separate analytics/AI service
10+	Scale out with multiple VMs or managed compute

CPU Driver Selected	Score	Reason
Base web/API application	1	FoodT race is a web-based cloud application.
Authentication and roles	1	Supports secure user login and role management.
API traffic	1	Handles REST API requests for QR scanning and tracking.
Database/search/reporting	2	Stores product records and generates reports.
Background jobs	1	Sends notifications and performs scheduled tasks.
Analytics/dashboard	2	Displays spoilage analytics and dashboard reports.
AI/Python module	3	Executes spoilage prediction using Python algorithms.
Total CPU Score	11	Sum of all selected scores.
Recommended vCPU	8 vCPU	Required to support analytics and AI processing efficiently

Activity 5: RAM Calibration

RAM must include operating system, runtime, application process, database/cache and a safety buffer.

RAM Formula

Estimated RAM = OS/runtime + app/API + database/cache + analytics/AI + 25% buffer.
If database is managed separately, do not include full database memory in the app VM.

Component	Guidance	Student Estimate
OS + runtime	1-2 GB	2 GB
Web/API service	1-2 GB	2 GB
Small database	2-4 GB	0GB
Medium database/cache	4-8 GB	4 GB
Analytics/reporting	4-8 GB extra	4 GB
AI/Python module	4-16 GB extra	8 GB
25% buffer	Subtotal x 0.25	5 GB
Final RAM	Round up to standard size	32 GB RAM

Activity 6: Storage Calibration

Storage must cover application files, database records, uploads, logs and backup/growth buffer.

Storage Item	Formula / Guidance	Student Estimate
OS and application	20-40 GB	40 GB
Database	Avg record KB x records/day x days x 1.3 / 1024 MB	30 GB
File uploads	Avg file MB x files/day x days x 1.3	40 GB
Logs	Requests/day x log KB x days / 1024 MB	10 GB
Backup/growth buffer	Subtotal x 30%	36 GB
Final storage	Round up to standard disk size	160 GB SSD

Technical Tip

Do not store large uploads permanently inside the VM disk when object storage is available. VM disk is for OS and application. Object storage is better for images, PDFs, media and documents.

Activity 7: Select the VM Plan

Plan	When It Fits	Student Choice
Single VM prototype	Small demo, simple app, low traffic	No
Two-tier design	App VM + managed/separate database	No
Three-tier design	Frontend, backend/API and database separated	Yes
Containerized backend	Microservices, portable API, CI/CD	Yes (Backend API)
Serverless function	Event-based task, notification, small automation	Yes (Notifications)
Managed database/storage	Reliability, backup and scaling required	Yes (Firebase Firestore & Cloud Storage)
Final VM Plan: selected plan, vCPU, RAM, storage and reason		
Selected Plan Three-tier Design Frontend → Backend → Database Reason Provides better scalability, security, and performance.		

Activity 8: Hypervisor and Virtualization Decision

Approach	Meaning	Best Use
Type 1 hypervisor	Runs directly on physical hardware	Production data center and cloud provider infrastructure
Type 2 hypervisor	Runs above an existing OS	Student lab, local testing, VirtualBox/VMware Workstation
Cloud-managed virtualization	Cloud provider hides hypervisor details	Most public cloud VM deployments

Container runtime	OS-level process isolation	Lightweight APIs, microservices, fast deployment
Context	Selected Approach	Justification
Local prototype/testing	Type-2 Hypervisor	VirtualBox
Cloud pilot deployment	Cloud-managed Virtualization	Firebase Cloud
Production growth scenario	Type-1 Hypervisor	Cloud infrastructure
Module suitable for container	Docker	Backend API

Activity 9: Snapshot and Clone Strategy

Snapshot protects a point in time. Clone creates a duplicate environment. Both are useful, but they are not the same.

Event	Snapshot or Clone?	Frequency / Timing	Retention / Reason
Before major deployment	Snapshot	Every Release	Keep for 30 days to enable rollback if deployment fails.
Before database schema change	Snapshot	Before every schema update	Keep for 30 days to restore the previous database version.
Before OS/package update	Snapshot	Before every system update	Keep for 15 days in case the update causes issues.
Before review/demo	Snapshot	Before each project review or demo	Keep until the review/demo is completed successfully.
Create testing environment	Clone	Before testing new features	Delete after testing to save cloud resources.
Create team demo copy	Clone	When a separate demo environment is required	Keep until the demonstration is completed, then remove.

Technical Tip

A snapshot is useful for rollback.
A clone is useful for parallel testing. A clone should not be treated as the only backup.

Activity 10: VM vs Container Decision

Criteria	Virtual Machine	Container
Isolation	Strong OS-level isolation	Process-level isolation
Startup	Slower	Faster
Resource use	Higher	Lower
Best for	Full OS, legacy app, database VM	Microservices, APIs, portable services

Output from CO2 AT1	How It Will Be Used Later
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Portability	Moderate	High
Student use	Clear infrastructure learning	Modern deployment learning
Capstone Module	VM / Container / Managed Service	Reason
Frontend	Container	Lightweight
Backend/API	Container	Easy deployment
Database	Managed Service	Automatic backup
File storage	Managed Service	Secure storage
AI/Python module	VM	Higher processing power
Analytics/dashboard	Container	Fast scaling
Notification/background jobs	Container	Background execution

Activity 11: Final Recommendation

Write the final infrastructure recommendation in 8-12 technical lines. Include VM size, hypervisor/virtualization approach, storage plan, snapshot/clone plan and VM-container decision.

Food Trace uses a **three-tier cloud architecture** with separate frontend, backend, and database services. The application is deployed using **8 vCPU, 32 GB RAM, and 160 GB SSD** to support QR tracking and spoilage prediction. Firebase Firestore and Cloud Storage provide secure, scalable data management. Docker containers are used for the frontend and backend services, while the AI/Python module runs on a dedicated virtual machine for better performance. Cloud-managed virtualization is selected for deployment, and snapshots are created before updates and demonstrations. Clones are used for testing and development environments. This infrastructure provides reliable performance, scalability, security, and efficient resource utilization for the Food Trace application.

Carry-Forward to CO2 AT2 and CO2 AT3

VM sizing values	Used as configuration target in CO2 AT2 practical evidence
Hypervisor/virtualization choice	Used to justify deployment environment
Snapshot/clone plan	Used as backup and testing evidence
VM/container comparison	Used in CO2 AT3 infrastructure design case
Final recommendation	Used as the capstone infrastructure baseline

GitHub Submission Checklist

- Completed workbook file.
- README with capstone title, sizing summary and final recommendation.
- Optional architecture diagram or table exported as image/PDF. • Repository link submitted in Google Classroom.